

# Aurora SVMAT Metric Formula Reference

Main body: v2/v3 research proxies and v4 physical aperture metrics. Appendix: legacy engineering metrics.

This document records the explicit mathematical definitions currently implemented in Aurora SVMAT Lab. The notation is aligned to the control-point sequence view  $c_i = \{\theta_i, w_i, M_i, z_i\}$ .

**RESEARCH USE ONLY. This document reflects the current code implementation and should be interpreted as a research reference rather than a clinical specification.**

## Notation

$i$  indexes adjacent control-point intervals from control point  $i$  to  $i+1$ .

$\theta_i$  is the gantry angle in degrees at control point  $i$ .

$z_i$  is the longitudinal or axial position in millimeters at control point  $i$ .

$w_i$  is the cumulative meterset weight at control point  $i$ .

For V2/V3/legacy only,  $A_i$  is the scalar aperture width surrogate at control point  $i$ , defined as the summed positive opening across both MLC layers.

$L_i$  is the total leaf-position vector at control point  $i$  across both layers.

$\Delta \theta_i = \text{wrapped}(\theta_{i+1} - \theta_i)$ , constrained to  $[-180, 180]$  degrees.

$\Delta z_i = z_{i+1} - z_i$ .

$\Delta w_i = w_{i+1} - w_i$ .

$\Delta A_i = |A_{i+1} - A_i|$ .

$\Delta L_i$  = summed absolute leaf travel between adjacent control points across both layers.

$CV(x) = \text{std\_pop}(x) / \text{mean}(x)$ , evaluated only on valid non-empty interval sets.

## Section 1. V2 Metrics

### longitudinal\_travel\_mm

#### Total absolute longitudinal travel

$\text{longitudinal\_travel\_mm} = \sum_i |\Delta z_i|$

Meaning: Total absolute longitudinal travel reconstructed from adjacent control-point isocenter  $z$  positions.

### total\_rotation\_deg

#### Total absolute gantry rotation

$\text{total\_rotation\_deg} = \sum_i |\Delta \theta_i|$

Meaning: Total absolute gantry rotation reconstructed from the observed angle sequence rather than declared direction tags.

## rotations

### Total gantry rotations

```
rotations = total_rotation_deg / 360
```

Meaning: Total gantry rotation expressed in full 360-degree turns.

## travel\_per\_rotation\_mm

### Longitudinal travel per full gantry rotation

```
travel_per_rotation_mm = longitudinal_travel_mm / rotations
```

Meaning: Average longitudinal travel per gantry rotation, derived from total travel and total rotation.

## projection\_pitch\_mean

### Mean projection pitch

```
projection_pitch_i = |delta z_i| / |delta theta_i|  
projection_pitch_mean = mean_i(projection_pitch_i)
```

Meaning: Mean projection pitch, defined as the mean of  $|\text{delta } z| / |\text{delta } \theta|$  over valid adjacent control-point intervals.

## projection\_pitch\_cv

### Projection pitch variability

```
projection_pitch_i = |delta z_i| / |delta theta_i|  
projection_pitch_cv = CV(projection_pitch_i)
```

Meaning: Coefficient of variation of projection pitch. Lower values mean steadier theta-z coupling.

## projection\_mu\_density\_mean\_proxy

### Mean projection meterset-weight density proxy

```
projection_mu_density_proxy_i = delta w_i / |delta z_i|  
projection_mu_density_mean_proxy = mean_i(projection_mu_density_proxy_i)
```

Meaning: Mean projection MU-density proxy, using cumulative meterset-weight change per millimeter of longitudinal travel.

## projection\_mu\_density\_cv\_proxy

### Variability of the projection meterset-weight density proxy

```
projection_mu_density_proxy_i = delta w_i / |delta z_i|  
projection_mu_density_cv_proxy = CV(projection_mu_density_proxy_i)
```

Meaning: Coefficient of variation of the projection MU-density proxy across adjacent control-point intervals.

## projection\_aperture\_change\_mean

### Mean aperture-change rate along the longitudinal axis

```
delta A_i = |A_(i+1) - A_i|  
projection_aperture_change_i = delta A_i / |delta z_i|  
projection_aperture_change_mean = mean_i(projection_aperture_change_i)
```

Meaning: Mean projection aperture-change rate, defined as absolute aperture-width change per millimeter of longitudinal travel.

## projection\_aperture\_change\_cv

### Variability of the aperture-change rate

```
delta A_i = |A_(i+1) - A_i|  
projection_aperture_change_i = delta A_i / |delta z_i|  
projection_aperture_change_cv = CV(projection_aperture_change_i)
```

Meaning: Coefficient of variation of the projection aperture-change rate.

### **projection\_leaf\_travel\_mean**

#### **Mean leaf-travel rate across both MLC layers**

```
delta L_i = sum_leaves |leaf_(i+1) - leaf_i| across MLCX1 and MLCX2
projection_leaf_travel_i = delta L_i / |delta z_i|
projection_leaf_travel_mean = mean_i(projection_leaf_travel_i)
```

Meaning: Mean projection leaf-travel rate, defined as summed leaf travel per millimeter of longitudinal travel.

### **projection\_leaf\_travel\_cv**

#### **Variability of the total leaf-travel rate**

```
delta L_i = sum_leaves |leaf_(i+1) - leaf_i| across MLCX1 and MLCX2
projection_leaf_travel_i = delta L_i / |delta z_i|
projection_leaf_travel_cv = CV(projection_leaf_travel_i)
```

Meaning: Coefficient of variation of the projection leaf-travel rate.

### **projection\_leaf\_travel\_mean\_mlcx1**

#### **Mean leaf-travel rate for MLCX1**

```
delta L_i^(MLCX1) = sum_x1 |x1_(i+1) - x1_i|
projection_leaf_travel_mlcx1_i = delta L_i^(MLCX1) / |delta z_i|
projection_leaf_travel_mean_mlcx1 = mean_i(projection_leaf_travel_mlcx1_i)
```

Meaning: Mean projection leaf-travel rate for the MLCX1 layer only, normalized by longitudinal travel.

### **projection\_leaf\_travel\_cv\_mlcx1**

#### **Variability of the MLCX1 leaf-travel rate**

```
delta L_i^(MLCX1) = sum_x1 |x1_(i+1) - x1_i|
projection_leaf_travel_mlcx1_i = delta L_i^(MLCX1) / |delta z_i|
projection_leaf_travel_cv_mlcx1 = CV(projection_leaf_travel_mlcx1_i)
```

Meaning: Coefficient of variation of the MLCX1 projection leaf-travel rate.

### **projection\_leaf\_travel\_mean\_mlcx2**

#### **Mean leaf-travel rate for MLCX2**

```
delta L_i^(MLCX2) = sum_x2 |x2_(i+1) - x2_i|
projection_leaf_travel_mlcx2_i = delta L_i^(MLCX2) / |delta z_i|
projection_leaf_travel_mean_mlcx2 = mean_i(projection_leaf_travel_mlcx2_i)
```

Meaning: Mean projection leaf-travel rate for the MLCX2 layer only, normalized by longitudinal travel.

### **projection\_leaf\_travel\_cv\_mlcx2**

#### **Variability of the MLCX2 leaf-travel rate**

```
delta L_i^(MLCX2) = sum_x2 |x2_(i+1) - x2_i|
projection_leaf_travel_mlcx2_i = delta L_i^(MLCX2) / |delta z_i|
projection_leaf_travel_cv_mlcx2 = CV(projection_leaf_travel_mlcx2_i)
```

Meaning: Coefficient of variation of the MLCX2 projection leaf-travel rate.

### **theta\_z\_coupling\_cv**

#### **Theta-z coupling variability**

```
theta_z_coupling_cv = projection_pitch_cv
```

Meaning: Theta-z coupling variability, reported here as the same projection-pitch coefficient of variation.

### **mu\_z\_coupling\_cv\_proxy**

#### **Meterset-weight-density to z coupling variability proxy**

`mu_z_coupling_cv_proxy = projection_mu_density_cv_proxy`

Meaning: MU-z coupling variability proxy, reported as the coefficient of variation of meterset-weight density over longitudinal travel.

### **mlc\_z\_coupling\_cv**

#### **MLC-z coupling variability across both layers**

`mlc_z_coupling_cv = projection_leaf_travel_cv`

Meaning: MLC-z coupling variability, reported here as the coefficient of variation of the projection leaf-travel rate.

### **mlcx1\_z\_coupling\_cv**

#### **MLCX1-z coupling variability**

`mlcx1_z_coupling_cv = projection_leaf_travel_cv_mlcx1`

Meaning: MLCX1-z coupling variability, reported as the coefficient of variation of the MLCX1 projection leaf-travel rate.

### **mlcx2\_z\_coupling\_cv**

#### **MLCX2-z coupling variability**

`mlcx2_z_coupling_cv = projection_leaf_travel_cv_mlcx2`

Meaning: MLCX2-z coupling variability, reported as the coefficient of variation of the MLCX2 projection leaf-travel rate.

## Section 2. V3 Research Metrics

These research-facing v3 metrics are the immediately implementable Aurora extensions derived directly from currently parsed RTPLAN control-point data.

### reversal\_symmetry\_index

#### Bidirectional modulation symmetry

mean over selected burden families of  $[1 - \text{abs}(\text{forward\_mean} - \text{backward\_mean}) / (\text{abs}(\text{forward\_mean}) + \text{abs}(\text{backward\_mean}))]$

Meaning: Symmetry score comparing forward and backward beam groups across matched modulation summaries. Higher values mean more balanced bidirectional modulation.

### forward\_backward\_metric\_difference

#### Forward-versus-backward modulation difference

$\text{abs}(\text{mean\_forward}(\text{coupled\_modulation\_index}) - \text{mean\_backward}(\text{coupled\_modulation\_index}))$

Meaning: Absolute difference in average modulation burden between forward-moving and backward-moving beam groups.

### beam\_pair\_balance\_index

#### Beam-pair modulation balance

$1 - \text{abs}(\text{sum\_forward}(\text{coupled\_modulation\_index}) - \text{sum\_backward}(\text{coupled\_modulation\_index})) / (\text{sum\_forward} + \text{sum\_backward})$

Meaning: Balance score comparing the total modulation burden assigned to forward and backward beam groups.

### small\_opening\_fraction

#### Fraction of narrow effective openings

$\text{weighted count}(\text{gap} < 10 \text{ mm and gap} > 0) / \text{weighted count}(\text{gap} > 0)$

Meaning: Weighted fraction of effective openings smaller than 10 mm.

### near\_closed\_fraction

#### Fraction of near-closed effective openings

$\text{weighted count}(\text{gap} < 2 \text{ mm and gap} > 0) / \text{weighted count}(\text{gap} > 0)$

Meaning: Weighted fraction of effective openings smaller than 2 mm.

### effective\_small\_gap\_burden

#### Weighted burden of narrow effective gaps

weighted mean of  $\max(0, 1 - \text{gap}/10 \text{ mm})$  over positive zipped-channel gap proxies;  $\text{abs}(\text{delta CMW})$  weights, zero increment falls back to 1

Meaning: Weighted severity score for narrow effective openings, with smaller openings contributing more burden.

### projection\_pitch\_p95

#### 95th percentile of projection pitch

nearest-rank 95th percentile of `projection_pitch_i`

Meaning: 95th percentile of projection pitch, using the nearest-rank interval summary.

### projection\_pitch\_max

#### Maximum projection pitch

`max(projection_pitch_i)`

Meaning: Maximum projection pitch observed across adjacent control-point intervals.

### **projection\_pitch\_top3\_mean**

#### **Mean of the top three projection-pitch intervals**

mean of the three largest `projection_pitch_i` values

Meaning: Mean of the three largest projection-pitch intervals, or all intervals if fewer than three exist.

### **projection\_mu\_density\_p95\_proxy**

#### **95th percentile of the projection meterset-weight density proxy**

nearest-rank 95th percentile of `projection_mu_density_proxy_i`

Meaning: 95th percentile of the projection MU-density proxy, using cumulative meterset-weight change per millimeter.

### **projection\_mu\_density\_max\_proxy**

#### **Maximum projection meterset-weight density proxy**

`max(projection_mu_density_proxy_i)`

Meaning: Maximum projection MU-density proxy observed across adjacent control-point intervals.

### **projection\_mu\_density\_top3\_mean\_proxy**

#### **Mean of the top three projection meterset-weight density proxy values**

mean of the three largest `projection_mu_density_proxy_i` values

Meaning: Mean of the three largest projection MU-density proxy values.

### **projection\_aperture\_change\_p95**

#### **95th percentile of the aperture-change rate**

nearest-rank 95th percentile of `projection_aperture_change_i`

Meaning: 95th percentile of the projection aperture-change rate.

### **projection\_aperture\_change\_max**

#### **Maximum aperture-change rate**

`max(projection_aperture_change_i)`

Meaning: Maximum projection aperture-change rate observed across adjacent control-point intervals.

### **projection\_aperture\_change\_top3\_mean**

#### **Mean of the top three aperture-change rates**

mean of the three largest `projection_aperture_change_i` values

Meaning: Mean of the three largest projection aperture-change rates.

### **projection\_leaf\_travel\_p95**

#### **95th percentile of the leaf-travel rate**

nearest-rank 95th percentile of `projection_leaf_travel_i`

Meaning: 95th percentile of the projection leaf-travel rate across both layers.

### **projection\_leaf\_travel\_max**

**Maximum leaf-travel rate**

```
max(projection_leaf_travel_i)
```

Meaning: Maximum projection leaf-travel rate across both layers.

**projection\_leaf\_travel\_top3\_mean****Mean of the top three leaf-travel rates**

```
mean of the three largest projection_leaf_travel_i values
```

Meaning: Mean of the three largest projection leaf-travel rates across both layers.

**head\_mu\_density\_mean\_proxy****Head-third mean MU-density proxy**

```
mean(projection_mu_density_proxy_i) over head-third intervals
```

Meaning: Mean projection MU-density proxy inside the head longitudinal third of the covered z-range.

**head\_mu\_density\_cv\_proxy****Head-third MU-density proxy variability**

```
CV(projection_mu_density_proxy_i) over head-third intervals
```

Meaning: Coefficient of variation of the projection MU-density proxy inside the head longitudinal third.

**head\_aperture\_change\_mean****Head-third mean aperture-change rate**

```
mean(projection_aperture_change_i) over head-third intervals
```

Meaning: Mean projection aperture-change rate inside the head longitudinal third of the covered z-range.

**head\_aperture\_change\_cv****Head-third aperture-change variability**

```
CV(projection_aperture_change_i) over head-third intervals
```

Meaning: Coefficient of variation of the projection aperture-change rate inside the head longitudinal third.

**head\_leaf\_travel\_mean****Head-third mean leaf-travel rate**

```
mean(projection_leaf_travel_i) over head-third intervals
```

Meaning: Mean projection leaf-travel rate inside the head longitudinal third of the covered z-range.

**head\_leaf\_travel\_cv****Head-third leaf-travel variability**

```
CV(projection_leaf_travel_i) over head-third intervals
```

Meaning: Coefficient of variation of the projection leaf-travel rate inside the head longitudinal third.

**mid\_mu\_density\_mean\_proxy****Mid-third mean MU-density proxy**

```
mean(projection_mu_density_proxy_i) over mid-third intervals
```

Meaning: Mean projection MU-density proxy inside the mid longitudinal third of the covered z-range.

### **mid\_mu\_density\_cv\_proxy**

#### **Mid-third MU-density proxy variability**

`CV(projection_mu_density_proxy_i) over mid-third intervals`

Meaning: Coefficient of variation of the projection MU-density proxy inside the mid longitudinal third.

### **mid\_aperture\_change\_mean**

#### **Mid-third mean aperture-change rate**

`mean(projection_aperture_change_i) over mid-third intervals`

Meaning: Mean projection aperture-change rate inside the mid longitudinal third of the covered z-range.

### **mid\_aperture\_change\_cv**

#### **Mid-third aperture-change variability**

`CV(projection_aperture_change_i) over mid-third intervals`

Meaning: Coefficient of variation of the projection aperture-change rate inside the mid longitudinal third.

### **mid\_leaf\_travel\_mean**

#### **Mid-third mean leaf-travel rate**

`mean(projection_leaf_travel_i) over mid-third intervals`

Meaning: Mean projection leaf-travel rate inside the mid longitudinal third of the covered z-range.

### **mid\_leaf\_travel\_cv**

#### **Mid-third leaf-travel variability**

`CV(projection_leaf_travel_i) over mid-third intervals`

Meaning: Coefficient of variation of the projection leaf-travel rate inside the mid longitudinal third.

### **tail\_mu\_density\_mean\_proxy**

#### **Tail-third mean MU-density proxy**

`mean(projection_mu_density_proxy_i) over tail-third intervals`

Meaning: Mean projection MU-density proxy inside the tail longitudinal third of the covered z-range.

### **tail\_mu\_density\_cv\_proxy**

#### **Tail-third MU-density proxy variability**

`CV(projection_mu_density_proxy_i) over tail-third intervals`

Meaning: Coefficient of variation of the projection MU-density proxy inside the tail longitudinal third.

### **tail\_aperture\_change\_mean**

#### **Tail-third mean aperture-change rate**

`mean(projection_aperture_change_i) over tail-third intervals`

Meaning: Mean projection aperture-change rate inside the tail longitudinal third of the covered z-range.

### **tail\_aperture\_change\_cv**

#### **Tail-third aperture-change variability**

`CV(projection_aperture_change_i) over tail-third intervals`

Meaning: Coefficient of variation of the projection aperture-change rate inside the tail longitudinal third.

### **tail\_leaf\_travel\_mean**

#### **Tail-third mean leaf-travel rate**

`mean(projection_leaf_travel_i) over tail-third intervals`

Meaning: Mean projection leaf-travel rate inside the tail longitudinal third of the covered z-range.

### **tail\_leaf\_travel\_cv**

#### **Tail-third leaf-travel variability**

`CV(projection_leaf_travel_i) over tail-third intervals`

Meaning: Coefficient of variation of the projection leaf-travel rate inside the tail longitudinal third.

### **layer\_imbalance\_index**

#### **Dual-layer travel imbalance**

`abs(mean(leaf_travel_i_MLCX1) - mean(leaf_travel_i_MLCX2)) / (abs(mean_MLCX1) + abs(mean_MLCX2))`

Meaning: Normalized difference between the mean MLCX1 and MLCX2 leaf-travel rates.

### **layer\_correlation\_index**

#### **Dual-layer travel correlation**

`correlation(projection_leaf_travel_i_MLCX1, projection_leaf_travel_i_MLCX2)`

Meaning: Correlation between the interval-by-interval MLCX1 and MLCX2 leaf-travel rates.

### **x1\_x2\_aperture\_disparity**

#### **Dual-layer aperture disparity surrogate**

`mean over control points of abs(sum|MLCX1| - sum|MLCX2|) / (sum|MLCX1| + sum|MLCX2|)`

Meaning: Surrogate side-opening disparity, computed from the relative difference in MLCX1 and MLCX2 position magnitudes.

### **x1\_x2\_leaf\_travel\_ratio**

#### **Dual-layer leaf-travel ratio**

`mean(projection_leaf_travel_i_MLCX1) / mean(projection_leaf_travel_i_MLCX2)`

Meaning: Ratio of mean MLCX1 leaf-travel rate to mean MLCX2 leaf-travel rate.

## Section 3. V4 Physical Aperture Metrics

Version: aurora-v4-physical-aperture-open-only. Real leaf boundaries define a common y-sub-strip grid. Each layer's position vector is split into A/B banks; intervals are intersected across layers and clipped to jaws. A and P are exact area and union perimeter. These seven descriptors are a physical-geometry extension with adapted MCS, not exact McNiven MCS or a reproduction of the old unit-width design. Only positive cumulative-weight increments contribute, represented by endpoint  $i+1$ . Weights are  $\Delta \text{CMW} / \text{beam final CMW} \times \text{BeamMeterset}$ . If any BeamMeterset is absent, every beam uses unit-normalized relative weights and a warning. Missing geometry or malformed/decreasing weights makes all seven unavailable. All seven use open-aperture conditional means: exclude area-zero endpoints and renormalize over retained weights. Closed leaves do not enter SAS or LSV. Exclusion count and weight fraction are disclosed in warnings; no open endpoint makes all seven unavailable. Endpoint sampling does not imply the whole interval was closed.

### mean\_ba

#### mean\_ba

`OpenMUmean(A_i); A_i = \sum_s h_{i,s} * g_{i,s}`, exact physical dual-layer intersection area

Meaning: Open-aperture conditional delivered-weight mean physical area after intersecting both MLC layers and clipping to X/Y jaws using real leaf boundaries; closed endpoints excluded and weights renormalized.

### mean\_bi

#### mean\_bi

`OpenMUmean(P_i^2 / (4 * \pi * A_i))`;  $P_i$  is the exact aperture-union perimeter;  $A_i = 0 \Rightarrow$  excluded

Meaning: Open-aperture conditional delivered-weight mean  $P^2 / (4 \pi A)$ , using exact physical union perimeter; closed endpoints excluded and weights renormalized; unavailable only when no valid open aperture remains.

### mean\_ca

#### mean\_ca

`OpenMUmean(P_i / A_i)`; units  $\text{mm}^{-1}$ ;  $A_i = 0 \Rightarrow$  excluded; this is not circularity

Meaning: Open-aperture conditional delivered-weight mean  $P/A$  in inverse millimeters, not circularity; closed endpoints excluded and weights renormalized.

### mean\_sas5

#### mean\_sas5

`OpenMUmean(count_s(0 < g_{i,s} < 5 mm) / count_s(g_{i,s} > 0))`; positive-height sub-strips only; closed  $\Rightarrow$  excluded

Meaning: Open-aperture conditional delivered-weight mean fraction of positive open physical sub-strips with gap strictly below 5 mm; equal sub-strip counts; closed leaves and fully closed endpoints excluded.

### mean\_sas10

#### mean\_sas10

`OpenMUmean(count_s(0 < g_{i,s} < 10 mm) / count_s(g_{i,s} > 0))`; positive-height sub-strips only; closed  $\Rightarrow$  excluded

Meaning: Open-aperture conditional delivered-weight mean fraction of positive open physical sub-strips with gap strictly below 10 mm; equal sub-strip counts; closed leaves and fully closed endpoints excluded.

## **mean\_mcs\_aurora**

### **mean\_mcs\_aurora**

```
OpenMUmean(AAV_i*LSV_i); AAV_i=A_i/sum_s max_positive_endpoint(h_i,s*g_i,s);  
LSV_i=1-sum_adj_open abs(delta g)/((n_open-1)*max_open(g)); n_open<=1 => LSV 1; zero  
envelope => MCS 0
```

Meaning: Open-aperture conditional delivered-weight mean AAV times positive-gap LSV. AAV uses beam maxima of clipped sub-strip areas. Closed endpoints excluded; weights renormalized. Adapted MCS, not exact McNiven MCS.

## **mcs\_complexity\_aurora**

### **mcs\_complexity\_aurora**

```
1-mean_mcs_aurora; complement of the physical-aperture adapted MCS
```

Meaning: One minus the Aurora physical-aperture adapted MCS; larger values indicate greater modulation complexity under this version-specific definition.

## Appendix A. Legacy Engineering Metrics

These legacy metrics are retained for engineering comparison with the first-pass Aurora prototype. They are not the preferred paper-style reporting set.

### axial\_travel\_mm

#### Legacy total axial travel

$$\text{axial\_travel\_mm} = \sum_i |\text{delta } z_i|$$

Meaning: Legacy engineering metric for total absolute axial travel reconstructed from adjacent control-point isocenter z positions.

### gantry\_rotation\_deg

#### Legacy total gantry rotation

$$\text{gantry\_rotation\_deg} = \sum_i |\text{delta } \theta_i|$$

Meaning: Legacy engineering metric for total absolute gantry rotation reconstructed from the observed angle sequence.

### mm\_per\_deg

#### Legacy axial travel per gantry degree

$$\text{mm\_per\_deg} = \text{axial\_travel\_mm} / \text{gantry\_rotation\_deg}$$

Meaning: Legacy engineering metric for total axial travel divided by total gantry rotation.

### mm\_per\_rotation

#### Legacy axial travel per full rotation

$$\text{mm\_per\_rotation} = \text{axial\_travel\_mm} / (\text{gantry\_rotation\_deg} / 360)$$

Meaning: Legacy engineering metric for total axial travel per full 360-degree gantry rotation.

### pitch\_consistency

#### Legacy pitch consistency metric

$$\begin{aligned} \text{interval\_pitch}_i &= |\text{delta } z_i| / |\text{delta } \theta_i| \\ \text{pitch\_consistency} &= \text{CV}(\text{interval\_pitch}_i) \end{aligned}$$

Meaning: Legacy engineering pitch variability metric, equivalent to the coefficient of variation of interval pitch.

### mu\_per\_mm

#### Legacy MU density along the axial axis

$$\text{mu\_per\_mm} = \text{total\_mu\_or\_weight\_proxy} / \text{axial\_travel\_mm}$$

Meaning: Legacy engineering MU density per millimeter of axial travel. Uses explicit total MU when available, otherwise cumulative meterset-weight span as a transparent proxy.

### aperture\_change\_per\_mm

#### Legacy aperture-change density

$$\text{aperture\_change\_per\_mm} = \sum_i \text{delta } A_i / \sum_i |\text{delta } z_i|$$

Meaning: Legacy engineering metric for total absolute aperture-width change normalized by total axial travel.

### leaf\_travel\_per\_mm

#### Legacy leaf-travel density

```
leaf_travel_per_mm = sum_i delta L_i / sum_i |delta z_i|
```

Meaning: Legacy engineering metric for total absolute leaf travel normalized by total axial travel.

### **coupled\_modulation\_index**

#### **Legacy engineering aggregate**

```
n(x) = x / (1 + x)
coupled_modulation_index = mean( n(aperture_change_per_mm), n(leaf_travel_per_mm),
n(mu_density_variability), n(pitch_consistency) )
```

Meaning: Legacy engineering aggregate of normalized aperture change, leaf travel, MU-density variability, and pitch variability.